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Title of the Project: Early Dropout Risk Prediction for CSE Students at BUBT Using Institutional Learning Management Data and Explainable ML

Introduction

The continuous tracking and dynamic prediction of student dropout in higher education institutions have emerged as fundamental pillars of modern educational data mining. To build effective early warning systems, researchers must analyze diverse behavioral and institutional data channels, balancing the trade-offs between complex black-box model accuracy and post-hoc model interpretability. Existing literature spans a wide range of approaches, including the collection of digital footprints from Learning Management Systems (LMS) like Moodle, historical academic records, and unstructured text-based feedback. This section provides a comprehensive, paragraph-by-paragraph evaluation of ten significant studies in this domain, identifying their core methodologies, experimental datasets, performance achievements, and underlying gaps or limitations.

Detailed Review

[1] predicted student dropout using activity logs from the Moodle platform. They applied the CatBoost model and used a balancing method called ADASYN to fix data problems. Their final model performed very well, achieving an F1-score of 0.86. The study found that long periods of student inactivity on Moodle was the biggest warning sign of dropout risk.

[2] Studied the problem of student dropout in the higher education system of Finland. In this research, the authors used both traditional academic data and Moodle (LMS) activity records. They selected main features like total completed credits, changes in GPA, and course failure history. For classification, they applied machine learning models such as CatBoost, Random Forest, and **XGBoost,** which achieved high accuracy. However, the major limitation is that these advanced models work like a **"black-box.**" They can predict who will drop out, but they do not explain the exact reasons for individual students. Additionally, since the study used data from a Western educational system, the findings may not directly apply to universities with different grading frameworks.

[3] predicted student dropout at a Hungarian university using interpretable machine learning on a dataset of 8,508 students. Using pre-enrollment data like high school grades, their CatBoost model achieved a 0.774 AUC score. They used SHAP and LIME to explain that high school GPA and math scores were the top predictors of student success.

[4] grouped at-risk students by analyzing Moodle activity logs. They used PCA and UMAP for data reduction, followed by clustering tools like BIRCH and DBSCAN to group students by engagement. Using Random Forest, they found that the BIRCH model worked best and showed that students with low online interaction had a dropout risk of over 61%

[5] Cho et al. (2023) predicted student dropout using a dataset of 20,050 students at a South Korean university. They compared Logistic Regression, Random Forest, SVM, and LightGBM models. To handle a low 5% dropout rate, they used SMOTE and ADASYN for data balancing. LightGBM performed best, achieving an F1-score of 0.840 in 1.1 seconds. However, oversampling reduced the performance of nearly all models. The main limitation was a large data overlap between dropout and regular students regarding grades and leave-of-absence semesters. This overlap caused the artificial oversampled data to act as noise, leading to more prediction errors.

[6] Phan et al. (2021) predicted student dropout by combining academic records with written student feedback. Using a dataset of 14,391 students from a French institution, they applied text tools like doc2vec and BERT along with Logistic Regression. Their results showed that adding student comments greatly improved accuracy, revealing that students drop out for various reasons like issues with teachers or course content. However, the study was limited to only one private business school in France and only analyzed course evaluations, ignoring other text sources like emails or forum posts.

[7] Vaarma and Li (2024) studied student dropout at a Finnish university with a dataset of 8,813 students. They tested CatBoost, Neural Networks, and Logistic Regression using academic records, demographics, and Moodle logs. Through a monthly analysis over 60 months, they found that earned credits, failed courses, and Moodle activity were the top predictors, where CatBoost gave the best result (AUC 0.853). However, the study was limited to just one Finnish university. Also, they only used two basic Moodle features (activity count and trend) and left out student survey data like motivation or satisfaction.

[8] Martínez and Castillo (2024) predicted student dropout using academic and demographic data from 2012 to 2023. They compared Gradient Boosted Tree, Random Forest, and Neural Networks across two approaches: using the first five semesters versus using only the first year (three semesters). They found that Gradient Boosted Tree performed best, reaching an accuracy of 0.954. Interestingly, looking only at the first year improved accuracy and saved computing power. However, the main limitation is that the dataset lacked socioeconomic details like family income, and the study was focused on just a single institution (UNADECA) in Central America.

[9] Handayani and Priyadi (2026) built a web-based early warning system to predict student academic performance at an Indonesian university. They used a dataset of 1,240 students from 2021 to 2024, mixing academic grades with Moodle LMS data like login frequency and submission timing. They compared Logistic Regression, Random Forest, Gradient Boosting, and LSTM models. Their results showed that Gradient Boosting performed best, with an accuracy of 89.1% and an AUC-ROC of 0.931. Also, SHAP analysis proved that LMS behavior, especially weekly logins, was a top predictor. Although 32 advisors gave the system a good usability score (78.4), the study was limited to just one institution. Additionally, it did not test whether the system's actual interventions improved student grades compared to a control group.

[10] Zerkouk et al. developed a model named SentiDrop to predict student dropout in distance learning. They combined socio-demographic and behavioral data with sentiment analysis of student comments, using BERT to process text and an ensemble of XGBoost, Random Forest, and Logistic Regression for predictions. Their findings showed that adding sentiment data raised accuracy to 84%, revealing that negative comments in the first month are strong warning signs of dropout. However, the study was limited to a single distance learning platform. Also, the model can produce many false positives, which might waste intervention resources on students who do not actually need help.

Research Gap and Our Contributions

- **The Transparency Gap:** While high-performing machine learning architectures introduced in studies like [1], [2], [5], and [8] achieve strong statistical boundaries, they operate strictly as traditional "black-box" configurations. They output high-risk categorical labels without providing individualized, transparent reasoning to guide academic advisors. Conversely, newer frameworks that attempt to integrate interpretability tools like SHAP either restrict their analysis completely to static pre-enrollment historical profiles [3]—leaving them blind to active semester behavioral dynamics—or rely on very limited dataset samples [9] without addressing intervention scaling.
- **The Feature & Algorithmic Noise Gap:** Unsupervised behavioral studies like [4] rely purely on grouping tendencies without assigning localized feature justifications, while traditional academic record pipelines [5] suffer from noticeable performance degradation because oversampling tools introduce heavy noise inside highly overlapping feature zones. Furthermore, recent advanced multimodal frameworks [6], [7], [8], [9], [10] remain heavily constrained by foreign geographical boundaries, lack structural socioeconomic variables, or extract overly basic, aggregate-level LMS features without granular temporal context.

We Will Contribute

To bridge these interconnected limitations, this research introduces a localized, high-resolution, and fully interpretable early warning system designed to improve student retention methodologies. The core contributions of our proposed system are outlined below:

1. **Multi-Modal Data Integration:** Unlike systems relying on single-source streams, our framework fuses longitudinal academic transcript trajectories (**CGPA progression**) with highly granular digital interaction footprints from **Moodle LMS logs** (specifically tracking weekly login frequency and assignment submission behaviors).
2. **Explainable AI (XAI) Framework:** We completely replace the operational limitations of traditional black-box models by embedding a **post-hoc SHAP (SHapley Additive exPlanations) analysis pipeline**. This generates real-time, student-level risk breakdowns, displaying the exact behavioral weights driving each risk alert for academic counselors.
3. **Temporal Early Warning & Noise Mitigation:** Our methodology leverages temporal modeling to catch student detachment at the earliest possible time-step during an ongoing semester. Additionally, we implement advanced feature engineering alongside **SMOTE architectures** optimized explicitly to mitigate the class-overlap noise that penalized past balancing attempts.
4. **Socioeconomic and Localized Alignment:** This project utilizes an empirical, real-world dataset extracted directly from the **Computer Science and Engineering (CSE) department of BUBT**. This makes our framework the first of its kind explicitly engineered for and tailored to the specific socioeconomic backgrounds, grading parameters, and behavioral characteristics defining private university students in Bangladesh.

Author(s) & Reference	Core Dataset & Source	Dataset Size	Primary Methods / Algorithms Tested	Best Performing Model	Maximum Performance Achieved	Imbalance Handling Method	Model Interpretability (XAI)	Core Architectural / Data Gaps
Marcolino et al. [1]	Moodle LMS interaction logs	Not stated	CatBoost	CatBoost	F1-Score: 0.86	ADASYN	None (Black-box)	Omits academic history and demographics;

								no local explainability.
Vaarma & Li [2]	Finnish Higher Ed. (Academic + Moodle)	8,813 students	CatBoost, Random Forest, XGBoost	Not explicitly isolated (Multiple high)	High Accuracy attained	Not mentioned	None (Black-box)	Completely black-box; limited generalizability to non-Western systems.
Nagy & Molontay [3]	Hungarian University Records	8,508 students	CatBoost	CatBoost	AUC: 0.774	Not mentioned	SHAP and LIME	Relies purely on pre-enrollment data; blind to real-time semester behavior.
Pecuchova & Drlik [4]	Moodle LMS Platform Traces	Not stated	PCA, UMAP, BIRCH, DBSCAN, Random Forest	BIRCH Clustering	Dropout Risk: >61% (Low engagement)	None (Clustering)	None (Unsupervised)	Lacks student-level feature explanations; omits academic data integration.

Cho et al. [5]	South Korean University Records	20,050 students	Logistic Regression, Random Forest, SVM, LightGBM	LightGBM	F1-Score: 0.840 (in 1.1s)	SMOTE and ADASYN	None	High class overlap generated oversampling noise; decreased specificity.
Phan et al. [6]	Private Business School, France	14,391 students	doc2vec, BERT, Logistic Regression	BERT + Logistic Regression	Text integration boosted accuracy	Not mentioned	None	Single private business school context; text restricted to course evaluations.
Vaarma & Li [7]	Finnish University Records	8,813 students	CatBoost, Neural Networks, Logistic Regression	CatBoost	AUC: 0.853	Not mentioned	None	Restricted to one university; extracts only 2 basic Moodle features.
Martínez & Castillo [8]	UNADECA, Central America	11-year history (2012-2023)	Gradient Boosted Tree, Random Forest, Neural Networks	Gradient Boosted Tree	Accuracy: 0.954	Not mentioned	None	Lacks socioeconomic parameters; focused on a single area; no LMS logs.

Handayani & Priyadi [9]	Indonesian University Platform	1,240 students	Logistic Regression, Random Forest, Gradient Boosting, LSTM	Gradient Boosting	Accuracy: 89.1%, AUC: 0.931	Not mentioned	SHAP	Small dataset scale; did not evaluate real intervention success against a control group.
Zerkouk et al. [10]	Distance Learning Environment (Téluq)	Not stated	BERT, XGBoost, Random Forest, Logistic Regression	SentiDrop Ensemble	Accuracy: 84%	Not mentioned	None (Sentiment only)	Distance learning restriction; high false-positive rates waste resources.
Proposed System	BUBT (CSE Department)	Institutional Dataset	CatBoost, Gradient Boosting architectures	To be evaluated	To be evaluated	SMOTE (Planned)	SHAP (Planned)	Specifically targets Bangladeshi private university context; minimizes oversampling noise.

Analytical Feature Matrix Comparison

S. No.	Core Analytical Features / System Traits	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]	[10]	Proposed System (BUBT)
1	Moodle LMS Log Feature Extraction	✔	✔	✖	✔	✖	✖	✔	✖	✔	✔	✔
2	Historical Transcript Analysis (CGPA)	✖	✔	✔	✖	✔	✔	✔	✔	✔	✔	✔
3	Transparent Explanations (SHAP/LIME)	✖	✖	✔	✖	✖	✖	✖	✖	✔	✖	✔
4	Advanced Class Imbalance Mitigation	✔	✖	✖	✖	✔	✖	✖	✖	✖	✖	✔

5	Multi-Source Hybrid Data Fusion	✖	✔	✖	✖	✖	✔	✔	✖	✔	✔	✔
6	Continuous Temporal / Weekly Analysis	✖	✖	✖	✖	✖	✖	✔	✖	✖	✖	✔
7	Fine-Grained Moodle Attribute Tracking	✖	✖	✖	✔	✖	✖	✖	✖	✔	✖	✔
8	Student-Level Risk Factor Assignment	✖	✖	✔	✖	✖	✖	✖	✖	✔	✖	✔
9	Multi-Model Architecture Evaluation	✖	✔	✖	✖	✔	✖	✔	✔	✔	✔	✔

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